

- ❖ *System development programs* are used for developing and executing application software. These are:
 - Language translators
 - Linkers
 - Debuggers
 - Editors

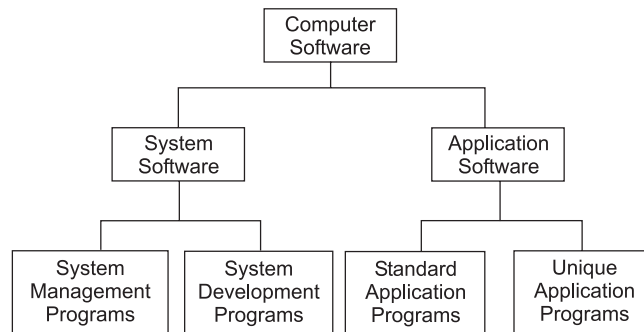


Fig. 1.15 Major categories of computer software

1.9.2 Application Software

Application software includes a variety of programs that are designed to meet the information processing needs of end users. They can be broadly classified into two groups:

- ❖ *Standard application programs* that are designed for performing common application jobs. Examples include:
 - Word processor
 - Spreadsheet
 - Database Manager
 - Desktop Publisher
 - Web Browser
- ❖ *Unique application programs* that are developed by the users themselves to support their specific needs. Examples include:
 - Managing the inventory of a store
 - Preparing pay-bills of employees in an organization
 - Reserving seats in trains or airlines

1.10 OVERVIEW OF OPERATING SYSTEM

LO 1.5

An operating system (OS) is a software that makes the computer hardware to work. While the hardware provides 'raw computer power', the OS is responsible for making the computer power useful for the users. OS is the main component of system software and therefore must be loaded and activated before we can accomplish any other task. The main functions include:

- ❖ Operates CPU of the computer.
- ❖ Controls input/output devices that provide the interface between the user and the computer.
- ❖ Handles the working of application programs with the hardware and other software systems.

- ❖ Manages the storage and retrieval of information using storage devices such as disks.

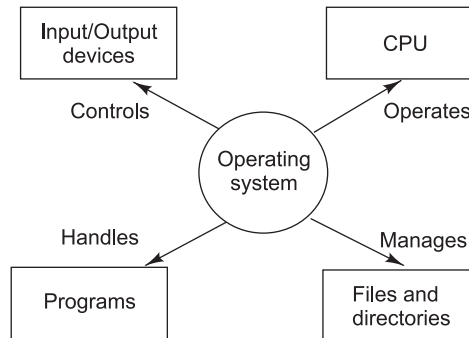


Fig. 1.16 The roles of an operating system

Based on their capabilities and the types of applications supported, the operating systems can be divided into the following six major categories:

- ❖ **Batch operating system** — This is the earliest operating system, where only one program is allowed to run at one time. We cannot modify any data used by the program while it is being run. If an error is encountered, it means starting the program from scratch all over again. A popular batch operating system is MS DOS.
- ❖ **Interactive operating system** — This operating system comes after the batch operating system, where also only one program can run at one time. However, here, modification and entry of data are allowed while the program is running. An example of an interactive operating system is Multics (Multiplexed Information and Computing Service).
- ❖ **Multiuser operating system** — A multiuser operating system allows more than one user to use a computer system either at the same time or at different times. Examples of multiuser operating systems include Linux and Windows 2000.
- ❖ **Multi-tasking operating system** — A multi-tasking operating system allows more than one program to run at the same time. Examples of multi-tasking operating systems include Unix and Windows 2000.
- ❖ **Multithreading operating system** — A multithreading operating system allows the running of different parts of a program at the same time. Examples of multithreading operating system include UNIX and Linux.
- ❖ **Real-time operating systems** — These operating systems are specially designed and developed for handling real-time applications or embedded applications. Example include MTOS, Lynx, RTX
- ❖ **Multiprocessor operating systems** — The multiprocessor operating system allows the use of multiple CPUs in a computer system for executing multiple processes at the same time. Example include Linux, Unix, Windows 7.
- ❖ **Embedded operating systems** — The embedded operating system is installed on an embedded computer system, which is primarily used for performing computational tasks in electronic devices. Example include Palm OS, Windows CE

1.10.1 MS DOS Operating System

MS DOS or Microsoft Disk Operating System, which is marketed by Microsoft Corporation and is one of the most commonly used members of the DOS family of operating systems. MS DOS is a command line

user interface, which was first introduced in 1981 for IBM computers. Although MS DOS, nowadays, is not used as a stand-alone product, but it comes as an integrated product with the various versions of Windows.

In MS DOS, unlike Graphical User Interface (GUI)-based operating systems, there is a command line interface, which is known as MS DOS prompt. Here, we need to type the various commands to perform the operations in MS DOS operating system. The MS DOS commands can be broadly categorized into the following three classes:

- ❖ **Environment command** — These commands usually provide information on or affects operating system environment. Some of these commands are:
 - **CLS**: It allows the user to clear the complete content of the screen leaving only the MS-DOS prompt.
 - **TIME**: It allows the user to view and edit the time of the computer.
 - **DATE**: It allows the user to view the current date as well as change the date to an alternate date.
 - **VER**: It allows us to view the version of the MS-DOS operating system.
- ❖ **File manipulation command** — These commands help in manipulating files, such as copying a file or deleting a file. Some of these commands include:
 - **COPY**: It allows the user to copy one or more files from one specified location to an alternate location.
 - **DEL**: It helps in deleting a file from the computer.
 - **TYPE**: It allows the user to view the contents of a file in the command prompt.
 - **DIR**: It allows the user to view the files available in the current and/or parent directories.
- ❖ **Utilities** — These are special commands that perform various useful functions, such as formatting a diskette or invoking the text editor in the command prompt. Some of these commands include:
 - **FORMAT**: It allows the user to erase all the content from a computer diskette or a fixed drive.
 - **EDIT**: It allows the user to view a computer file in the command prompt, create and modify the computer files.

1.10.2 MS Windows Operating System

Windows Architecture

The architecture of Windows operating system comprises a modular structure that is compatible with a variety of hardware platforms. Figure 1.17 shows the architecture of Windows 2000; the later releases of Windows operating systems are based on similar architecture.

At a high level, the architecture is divided into three layers, viz.

- ❖ **User mode**: Comprises application and I/O specific software components
- ❖ **Kernel mode**: Has complete access to system resources and hardware
- ❖ **Hardware**: Comprises underlying hardware platform

User Mode

The various subsystems in the user mode are divided into the following two categories:

- ❖ **Environment subsystems**: Comprise subsystems that run applications written for other operating systems. These subsystems cannot directly request hardware access; instead such requests are processed by virtual memory manager present in the kernel mode. The three main environment subsystems include Win32, OS/2 and POSIX. Each of these subsystems possess dynamic link libraries for converting user application calls to Windows calls.
- ❖ **Integral subsystems**: Takes care of the operating system specific functions on behalf of the environment subsystems. The various integral subsystems include workstation service, server service and security.

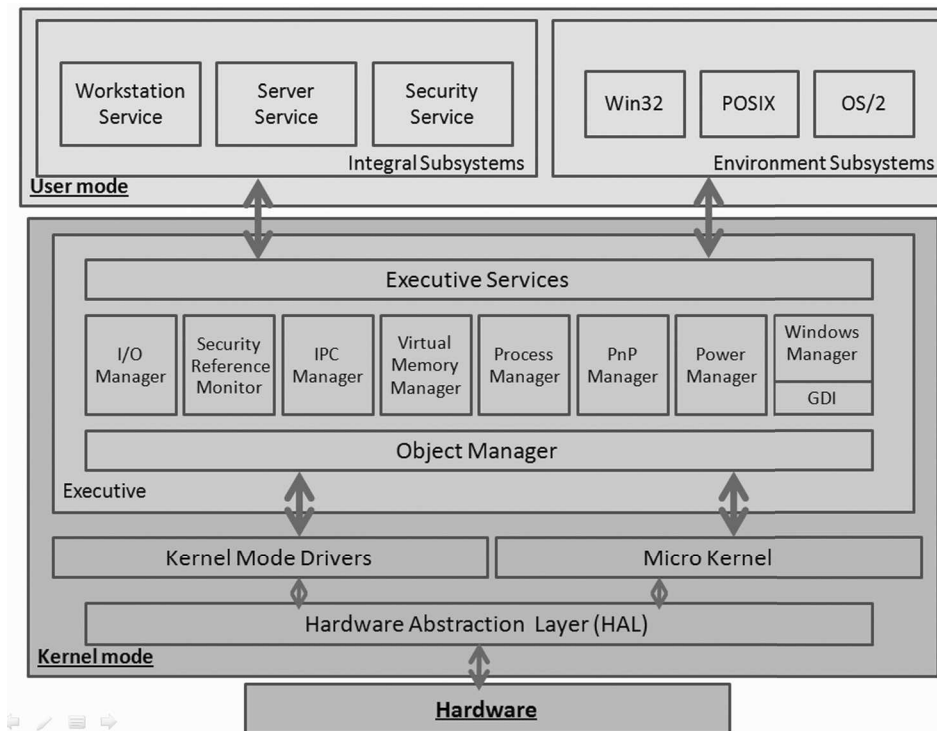


Fig. 1.17 The architecture of Windows 2000

Kernel Mode

The kernel mode comprises various components with each component managing specific system function. Each of the components is independent and can be removed, upgraded or replaced without rewriting the entire system. The various kernel-mode components include:

- ❖ **Executive:** Comprises the core operating system services including memory management, process management, security, I/O, inter process communication etc.
- ❖ **Kernel:** Comprises the core components that help in performing fundamental operating system operations including thread scheduling, exception handling, interrupt handling, multiprocessor synchronization, etc.
- ❖ **HAL:** Acts as a bridge between generic hardware communications and those specific to the underlying hardware platform. It helps in presenting a consistent view of system bus, DMA, interrupt controllers, timers, etc. to the kernel.
- ❖ **I/O manager:** Handles requests for accessing I/O devices by interacting with the relevant device drivers.
- ❖ **Security reference monitor:** Performs access validation and audit checks for Windows objects including files, processes, I/O devices, etc.
- ❖ **Virtual Memory Manager:** Performs virtual memory management by mapping virtual addresses to actual physical pages in computer's memory.
- ❖ **Process Manager:** Creates and deletes objects and threads throughout the life cycle of a process.

- ❖ **PnP manager:** Supports plug-and-play devices by determining the correct driver for a device and further loading the driver.
- ❖ **Power manager:** Performs power management for the various devices. It also optimizes power utilization by putting the devices to sleep that are not in use.
- ❖ **GDI:** Stands for Graphics Device Interface and is responsible for representing graphical objects in Windows environment. It also transfers the graphical objects to the output devices such as printer and monitor.
- ❖ **Object manager:** Manages Windows Executive objects and abstract data types that represent the various resources such as processes, threads, etc.

1.10.3 Unix Operating System

UNIX operating system was developed by a group of AT&T employees at Bell Labs in the year 1969. UNIX is primarily designed to allow multiple users access the computer at the same time and share resources. The UNIX operating system is written in C language. The significant properties of UNIX include:

- ❖ Multi-user capability
- ❖ Multi-tasking capability
- ❖ Portability
- ❖ Flexibility
- ❖ Security

Architecture of UNIX

UNIX has a hierarchical architecture consisting of several layers, where each layer provides a unique function as well as maintains interaction with its lower layers. The layers of the UNIX operating system are:

- ❖ Kernel
- ❖ Service
- ❖ Shell
- ❖ User applications

Figure 1.18 shows the various layers of the UNIX operating system.

- ❖ **Kernel** Kernel is the core of the UNIX operating system and it gets loaded into memory whenever we switch on the computer. Three components of kernel are:

- **Scheduler** — It allows scheduling the processing of various jobs.
- **Device driver** — It helps in controlling the Input/Output devices attached to the computer.
- **I/O buffer** — It controls the I/O operations in the computer.

Various functions performed by the kernel are:

- Initiating and executing different programs at the same time
- Allocating memory to various user and system processes
- Monitoring the files that reside on the disk
- Sending and receiving information to and from the network

- ❖ **Service** In the service layer, requests are received from the shell and they are then transformed into commands to the kernel. The service layer, which is also known as the *resident module layer*, is indistinguishable from the kernel and consists of a collection of programs providing various services, which include:

- Providing access to various I/O devices, such as keyboard and monitor
- Providing access to storage devices, such as disk drives
- Controlling different file manipulation activities, such as reading from a file and writing to a file

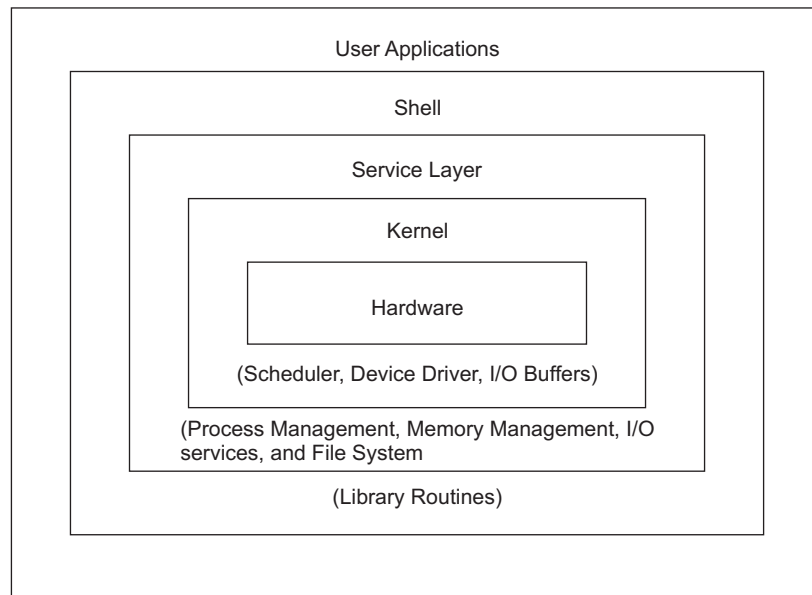


Fig. 1.18 The layers of UNIX operating system

- ❖ **Shell** The third layer in the UNIX architecture is the shell, which acts as an interface between a user and the computer for accepting the requests and executing programs. The shell is also known as the command interpreter that helps in controlling the interaction with the UNIX operating system. The primary function of the shell is to read the data and instructions from the terminal, and then execute commands and finally display the output on the monitor. The shell is also termed as the utility layer as it contains various library routines for executing routine tasks. The various shells that are found in the UNIX operating system are:
 - Bourne shell
 - C shell
 - Korn shell
 - Restricted shell
- ❖ **User applications** The last layer in the UNIX architecture is the user applications, which are used to perform several tasks and communicating with other users of UNIX. Some of the important examples of user applications include text processing, software development, database management and electronic communication.

1.11 MS WORD

LO 1.6

MS Word is application software that can be used to create, edit, save and print personal as well as professional documents in a very simple and efficient manner. MS Word is an important tool of the MS office suite that is mainly designed for word processing. Other word processing applications available are, Open Office Writer and Google Docs.